

Fundamental particles, forces and the Standard Model

Physicists have identified 13 building blocks that are the fundamental constituents of matter. Our everyday world is made of just three of these building blocks: the up quark, the down quark and the electron. This set of particles is all that's needed to make protons and neutrons and to form atoms and molecules. The electron neutrino, observed in the decay of other particles, completes the first set of four building blocks. For some reason nature has elected to replicate this first generation of quarks and leptons to produce a total of six quarks and six leptons, with increasing mass.

Like all quarks, the sixth quark, named top, is much smaller than a proton (in fact, no one knows how small quarks are), but the top is as heavy as a gold atom! Although there are reasons to believe that there are no more sets of quarks and leptons, theorists speculate that there may be other types of building blocks, which may partly account for the dark matter implied by astrophysical observations. This poorly understood matter exerts gravitational forces and manipulates galaxies. It will take Earth-based accelerator experiments to identify its fabric.

Scientists distinguish four elementary types of forces acting among particles: strong, weak, electromagnetic and gravitational force. The strong force is responsible for quarks "sticking" together to form protons, neutrons and related particles. The electromagnetic force binds electrons to atomic nuclei (clusters of protons and neutrons) to form atoms. The weak force facilitates the decay of heavy particles into smaller siblings. The gravitational force acts between massive objects. Although it plays no role at the microscopic level, it is the dominant force in our everyday life and throughout the universe.

Particles transmit forces among each other by exchanging force-carrying particles called bosons. These force mediators carry discrete amounts of energy, called quanta, from one particle to another. You could think of the energy transfer due to boson exchange as something like the passing of a basketball between two players. Each force has its own characteristic bosons: The gluon mediates the strong force; it "glues" quarks together. The photon carries the electromagnetic force; it also transmits light. The W and Z bosons represent the weak force; they introduce different types of decays.

Physicists expect that the gravitational force may also be associated with a boson particle. Named the graviton, this hypothetical boson is extremely hard to observe since, at the subatomic level, the gravitational force is many orders of magnitude weaker than the other three elementary forces. The Higgs boson is a particle associated with the Higgs field, the mechanism through which elementary particles gain mass. Without the Higgs field, or something similar, atoms would not form, and there would be no chemistry, no biology and no life.

The Higgs field is like a giant vat of molasses spread throughout the universe. Most of the known types of particles that travel through it stick to the molasses, which slows them down and makes them heavier. The Higgs boson is a particle that helps transmit the mass-giving Higgs force field, similar to the way a particle of light, the photon, transmits the electromagnetic field. The ATLAS and CMS experiments at CERN's Large Hadron Collider in Geneva, Switzerland, announced the discovery of the Higgs particle in July 2012. Although it is a staple of science fiction, antimatter is as real as matter.

For every particle, physicists have discovered a corresponding antiparticle, which looks and behaves in almost the same way. Antiparticles, though, have the opposite properties of their corresponding particles. An antiproton, for example, has a negative electric charge while a proton is positively charged. In the mid-1990s, physicists at CERN (1995) and Fermilab (1996) created the first anti-atoms. To learn more about the properties of the antimatter world, they carefully added a positron (the antiparticle of an electron) to an antiproton. The result: antihydrogen. Storing antimatter is a difficult task.

As soon as an antiparticle and a particle meet, they annihilate, disappearing in a flash of energy. Using electromagnetic force fields, physicists are able to store antimatter inside vacuum vessels for a limited amount of time. No one has ever directly observed dark matter, but two clues led astronomers to suspect its existence. First, when researchers measured the masses of all the stars and planets that make up galaxies, they discovered that the gravity of those objects alone would not be great enough to hold them

together.

Something they could not see must have been contributing mass and therefore gravitational pull. Second, they observed in space the kind of distortions of light usually caused by large masses in areas that seemed empty. The composition of dark matter is unknown, and its existence shows that the Standard Model of particle physics is incomplete. Several theories of particle physics, such as supersymmetry, predict that weakly interacting massive particles, WIMPs, exist with properties suitable for explaining dark matter. In the 20th century, astronomers first discovered that the universe was getting bigger.

They found this by observing something similar to the Doppler effect in the light coming from distant galaxies. The Doppler effect is what causes a car horn to change in pitch from high to low as it approaches and passes. This happens because the sound waves are compressed as the car moves toward you, resulting in a higher pitch, and are stretched as it recedes, resulting in a lower pitch. As an object approaches you, the light waves coming from it compress. Astronomers call this blueshift. When light waves stretch as an object moves farther away, astronomers call it redshift.

By measuring the spectrum of an astronomical object, astronomers can tell how much the space between the object and observer has stretched as the light traveled through it. When astronomer Vesto Slipher measured light coming from other galaxies, he found that almost all were redshifted, or moving away. He found that those that seemed dimmer and farther away had even higher redshifts. The universe was expanding. This led astronomers to the idea of the big bang. Astronomers assumed, however, that the force of gravity from all of the matter in the universe would slow the expansion.

They were in for a surprise in 1998 when they discovered that the expansion was actually speeding up. Astronomers discovered this when they measured the brightness of the light coming from a certain type of supernova that always explodes with roughly the same energy. The dimmer the light from the supernova, the farther the distance it had traveled to Earth. They discovered that the supernovae were farther away than their redshift measurements predicted. The universe was expanding at an accelerating rate.

Some particle astrophysicists think this is happening because a force with a repulsive gravity is pushing the universe apart. They call this force dark energy. Physicists call the theoretical framework that describes the interactions between elementary building blocks (quarks and leptons) and the force carriers (bosons) the Standard Model. Gravity is not yet part of this framework, and a central question of 21st-century particle physics is the search for a quantum formulation of gravity that could be included in the Standard Model.

Though still called a model, the Standard Model is a fundamental and well-tested physics theory. Physicists use it to explain and calculate a vast variety of particle interactions and quantum phenomena. High-precision experiments have repeatedly verified subtle effects predicted by the Standard Model. So far, the biggest success of the Standard Model is the unification of the electromagnetic and the weak forces into the so-called electroweak force. The consolidation is a milestone comparable to the unification of the electric and the magnetic forces into a single electromagnetic theory by J.C.

Maxwell in the 19th century. Physicists think it is possible to describe all forces with a Grand Unified Theory.

Particle physics: introductory concepts

What is particle physics?

We can define particle physics as the science of the really small things that make up the Universe. This includes leptons (such as the electron), the quarks that make up protons and neutrons, and gauge bosons, which mediate forces between particles. Particle physics is the study of fundamental subatomic particles, or elementary particles, including both matter (and antimatter) particles and the carrier particles of the fundamental interactions. By studying fundamental interactions, physicists try to comprehend how these elementary particles behave alone and with each other.

Besides studying elementary particles themselves, particle physicists also study the fundamental forces that govern how particles interact and decay. All the known forces of Nature can be traced to these fundamental forces: gravity, weak force, strong force and electromagnetic force. Our best understanding of particle physics is encapsulated in the Standard Model of Particle Physics. This can be compared to the Periodic Table - but for particles and forces. Physicists are constantly testing the limits of the Standard Model, looking for new additions or corrections to this theory.

Truth is, science is always working! And one day we might find that elementary particles aren't so elementary after all. The Standard Model is great at explaining a lot about the Universe, but there is still much that particle physicists do not understand. For example, only three out of four fundamental forces are described by the Standard Model, yet we do not know how gravity fits in? Why are we made of matter and not antimatter? These are some of the main motivators for the research carried out at ATLAS.

Particle Physics is sometimes called "high-energy physics" as physicists need to smash particles together at very high energies in enormous colliders to create and study some of these particles!

What is an elementary particle?

An elementary particle is not known to be made up of smaller entities. Elementary particles are the fundamental building blocks of the Universe. Quarks and electrons, for example, are elementary particles. Although both are elementary particles, electrons and quarks differ in many aspects. Whereas quarks together form nucleons (protons and neutrons) within the atomic nucleus, the electrons make up the shell of atoms. An atom may be tiny, but the diameter of the nucleus is ten thousand times smaller than that of the atom. Quarks and electrons are at least ten thousand times smaller than that!

What is the Large Hadron Collider?

The Large Hadron Collider (LHC) is a 27-km circular particle accelerator based at CERN in Geneva, Switzerland. It both accelerates and collides particles at record-breaking high energy in order to explore areas not charted by the Standard Model and search for new phenomena. The LHC is the final link in a chain of accelerators with increasingly higher energies. Each injects the beam into the next one, which brings the beam of particles to an even higher energy. Particles in the LHC are accelerated almost to the speed of light.

All particle accelerators usually have: - A source that generates electrically charged particles; - An electric field to accelerate the particles; - A magnetic field to control the paths of the particles. The LHC has the shape of a circle, meaning that the particles repeat the same circuit for as long as necessary, getting an energy boost at each turn. Inside the LHC, the particle beams travel through vacuum pipes - containers with as little residual air as possible - to avoid unwanted collisions with air molecules.

Beams can be made of different kinds of particles; the LHC uses mainly proton beams and sometimes nuclei of heavy atoms! After accelerating and smashing particles together, physicists detect and study the results of these collisions. Particle experiments are the enormous microscopes that detect the results of these collisions through the traces they leave behind.

How do LHC experiments work?

The Large Hadron Collider (LHC) accelerates two sets of particles to almost the speed of light (travelling in opposite directions), before setting them on a path to collide with each other. When they are smashed together, new particles are created, which can be detected and studied by experiments. ATLAS is the largest experiment on the LHC, and like most particle-physics detectors, it is made of different layers. Each of these layers play a unique role and is optimised for different particles - for example, tracking detectors are designed to trace the paths of electrically charged particles.

The ATLAS experiment sees up to 1.7 billion collisions every second and only a fraction of them can be recorded and stored. To make sure only the most interesting collisions are recorded, ATLAS uses super fast electronics and a special selection system. This decision-making process needs to be near-instantaneous and all the massive amounts of data quickly stored. ATLAS records over 10,000 TB of data per year - that's equivalent to 320,000 hours of 4K streaming!

What are particle decays?

When particles transform into multiple other particles, we call it a "particle decay". Particle decays are mediated by fundamental forces. Most of the interesting and relevant particles created in LHC collisions decay into other particles before they can be detected. The only way that physicists can study the properties of such particles is by measuring their decay products. A particle can decay in multiple different ways - and each of these has an associated probability. Find below an example of a particle decay where a Higgs boson decays into two Z bosons.

These Z bosons then decay into two electrons and two muons, respectively. These decays are commonly studied at the ATLAS experiment.

What is the Standard Model?

The Standard Model is the result of many decades of research and summarises the current knowledge scientists have on the fundamental forces and constituents of matter. The Standard Model describes all elementary particles physicists have found so far! Particles are grouped into matter particles (quarks and leptons) and force-carrying particles (bosons). Both quarks and leptons exist in three distinct sets, called families or generations. The Standard Model has been in development for quite some time. Even though the Standard Model is incomplete, it does an outstanding job at describing the known Universe.

Physicists are constantly testing the limits of the Standard Model, looking for new data to prove or refute it.

What about particle mass?

Weight is the measure of the force of gravity on an object. Mass is a measurement of how much matter is in an object. The mass of an object will never change, but the weight of an item can change based on its location. For example, in outer space you would have a different weight due to gravity being different. However, you will always have the same mass on Earth as you have in outer space! For very tiny particles mass and energy are equivalent! Particle mass is expressed in eV/c^2 , where "eV" represents energy (in electronvolts) and "c" is the speed of light.

The elementary particles we know have very different masses that can span several orders of magnitude. For example, neutrinos have really small masses, just a few eV, while others have high masses, on the order of 1,000,000,000 eV (1 GeV). Most elementary particles have masses on the order of 1,000,000 eV (1 MeV). The heaviest elementary particle is the top quark, that is as massive as an atom of gold (that has 197 nucleons).

What are particle generations?

Quarks and leptons are organised into generations. Each generation consists of a pair of a so-called up-type and a down-type quark (electric charges $+\frac{2}{3}$ and $-\frac{1}{3}$) and a pair of a neutral and a charged lepton (electric charges 0 and -1). The generations are organised by increasing mass. All visible matter in the Universe is made from the first generation of matter particles - up quarks, down quarks, and

electrons. This is because all second and third generation particles are unstable and quickly decay into first-generation particles. Physicists don't know why there are three generations.

It could be possible to have more, who knows!

Colour charge and antiparticles

Elementary particles are further organised following other properties, like electric charge. A property that is not as well-known outside particle physics is colour charge. There are three so-called colour charges: red, green, and blue; although there is no connection between the colour charge of quarks and gluons and the regular colours we know, it is just a name. Each quark carries a single colour charge, while gluons carry both a colour and an anticolour charge!

Protons and neutrons contain three quarks, each with one of three possible values of colour (red, blue, and green), turning protons and neutrons into a "colourless" object. For every type of matter particle we've found, there also exists a corresponding antimatter particle, or antiparticle. Antiparticles look and behave just like their corresponding matter particles, except they have opposite charges. When a particle and their antiparticle meet, they annihilate each other and create energy. When the Universe was formed after the Big Bang, matter and antimatter were produced in equal amounts.

However, from what scientists have observed so far, our Universe is made only of matter. Why is this? Could antiparticles have other properties that lead to their disappearance? The LHC could help to provide an answer. Antiparticles can be created in particle interactions, such as those in particle accelerators like the LHC or in the Earth's atmosphere by cosmic rays (see page 12).

Antiparticles in medicine

Antiparticles also have many uses in medicine! For example, a special kind of scanner called the PET (Positron Emission Tomography) uses positrons - the antiparticles of electrons - to look into the human body. During a PET scan, patients are injected with a tracer that produces positrons inside the body. When positrons bump into nearby electrons, they create a pair of photons that can be detected by the PET scanner. As the tracer collects in areas of high metabolic activity, the scan image can help doctors pinpoint areas of interest.

What are cosmic rays?

Cosmic Rays are high-energy subatomic particles that originate from outer space, mostly from supernovas and distant galaxies. Upon reaching the Earth, most cosmic rays are deflected by its magnetic field while the rest interact with the atmosphere and decay. Some of the products of these decays reach the Earth's surface. Cosmic rays are mainly high-energy protons. They collide with the nuclei of atoms in the Earth's upper atmosphere - like oxygen and nitrogen - creating more particles.

Most of these particles then decay and are stopped in the atmosphere before reaching the Earth's surface, with some exceptions like cosmic-ray muons. Muons do not interact strongly with matter, hence those produced in the atmosphere can penetrate even deep into the ground (and be measured by experiments like ATLAS). The rate of cosmic-ray muons arriving at the surface of the Earth is such that about 25 pass through your hand each minute! Cosmic rays constantly rain down on Earth but by the time they reach us it becomes difficult to trace where they have come from.

Cosmic rays and their interactions with the earth's atmosphere can be considered Nature's LHC. By using particle accelerators, physicists can mimic the cosmic ray collisions in the lab, though at much lower energies. But even though accelerators came to provide the best hunting ground for new particles, the physics of cosmic rays is still widely studied.

Gravity and open questions

While in the colouring books you read about the fundamental forces that operate between fundamental particles, you may have noticed that one is missing: gravity. Why is it not in the book? Well, that's because we cannot describe it at very small scales! Gravity is a force of attraction that acts between

everything with mass. On Earth, the gravitational force is what pulls everything towards the ground. It is responsible for the formation of stars and galaxies. It has influenced the way that the Universe has evolved since its origin in the Big Bang.

As elementary particles have mass, they are also affected by gravity. Scientists are working on understanding how gravity can fit inside the Standard Model. Like the other forces, it is hypothesised that the gravitational force could be carried by a suitable messenger particle: the graviton! Will scientists find it at ATLAS? We already know that the Standard Model cannot explain everything we observe in the Universe. For example, how does gravity fit in it? What is dark matter and what happened to antimatter?

Scientists have come up with a lot of different explanations for these unresolved mysteries that go beyond the Standard Model. There could be new particles, forces, states of matter, or particle-like phenomena - all of which may be discovered by ATLAS. Some new theories - like Supersymmetry - extend the mathematical framework of the Standard Model. Supersymmetry doubles the number of fundamental particles, stating that each particle of the Standard Model has a supersymmetric “partner” particle.

Searches by ATLAS physicists can target specific theories - for example, looking for a supersymmetric top quark - or can look broadly for evidence of new phenomena - for example, looking into particle decay products to see if there is a non-Standard Model particle that decayed into Standard Model particles.

The Standard Model

The Standard Model of Particle Physics is scientists' current best theory to describe the most basic building blocks of the universe. It explains how particles called quarks (which make up protons and neutrons) and leptons (which include electrons) make up all known matter. It also explains how force-carrying particles, which belong to a broader group of bosons, influence the quarks and leptons. The Standard Model explains three of the four fundamental forces that govern the universe: electromagnetism, the strong force, and the weak force.

Electromagnetism is carried by photons and involves the interaction of electric fields and magnetic fields. The strong force, which is carried by gluons, binds together atomic nuclei to make them stable. The weak force, carried by W and Z bosons, causes nuclear reactions that have powered our Sun and other stars for billions of years. The fourth fundamental force is gravity, which is not adequately explained by the Standard Model. Despite its success at explaining the universe, the Standard Model does have limits.

For example, the Higgs boson gives mass to quarks, charged leptons (like electrons), and the W and Z bosons. However, we do not yet know whether the Higgs boson also gives mass to neutrinos - ghostly particles that interact very rarely with other matter in the universe. Also, physicists understand that about 95 percent of the universe is not made of ordinary matter as we know it. Instead, much of the universe consists of dark matter and dark energy that do not fit into the Standard Model. DOE has a long history of supporting research into fundamental particles.

Five of the six types of quarks, one type of lepton, and all three neutrinos were discovered at what are now DOE national laboratories. Researchers supported by the DOE Office of Science, often in collaboration with scientists from around the world, have contributed to Nobel Prize-winning discoveries and measurements that refined the Standard Model. These efforts continue today, with experiments that make precision tests of the Standard Model and further improve measurements of particle properties and their interactions.

Theorists work with experimental scientists to develop new avenues to explore the Standard Model. This research may also provide insight into what sorts of unknown particles and forces might explain dark matter and dark energy as well as explain what happened to antimatter after the big bang. All ordinary matter, including every atom on the periodic table of elements, consists of only three types of matter particles: up and down quarks, which make up the protons and neutrons in the nucleus, and electrons that surround the nucleus. The complete Standard Model took a long time to build. Physicist J.J.

Thomson discovered the electron in 1897, and scientists at the Large Hadron Collider found the final piece of the puzzle, the Higgs boson, in 2012.

Quarks and gluons

Quarks and gluons are the building blocks of protons and neutrons, which in turn are the building blocks of atomic nuclei. Scientists' current understanding is that quarks and gluons are indivisible - they cannot be broken down into smaller components. They are the only fundamental particles to have something called color-charge. Quarks can have a positive or negative electric charge (like protons and electrons). Gluons have no electric charge. Both quarks and gluons have three additional states of charge: positive and negative redness, greenness, and blueness.

These so-called color charges are just names - they are not related to actual colors. The force that connects positive and negative color charges is called the strong nuclear force. This strong nuclear force is the most powerful force involved with holding matter together. It is much stronger than the three other fundamental forces: gravity, electromagnetism, and the weak nuclear forces. Because the strong nuclear force is so powerful, it makes it extremely difficult to separate quarks and gluons. Because of this, quarks and gluons are bound inside composite particles.

The only way to separate these particles is to create a state of matter known as quark-gluon plasma. In this plasma, the density and temperature are so high that protons and neutrons melt. This soup of quarks and gluons permeated the entire universe until a few fractions of a second after the Big Bang, when the universe cooled enough that quarks and gluons froze into protons and neutrons. Today, scientists study this quark-gluon plasma at special facilities such as the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratory.

DOE supports research on the interaction of quarks and gluons, the ways they combine into composite particles called hadrons, and the way they behave at high temperature and density. Scientists study these topics at DOE accelerator facilities like RHIC and the Continuous Electron Beam Accelerator Facility (CEBAF) at the Thomas Jefferson National Accelerator Facility. The theory that describes the strong nuclear force known as Quantum-Chromodynamics is notoriously difficult to solve. However, it can be simulated on supercomputers built and maintained at DOE facilities.

DOE has been a leader in the study of quarks and gluons since the 1960s. The idea of quarks was proposed in 1964, and evidence of their existence was seen in experiments in 1968 at the Stanford Linear Accelerator Center (SLAC). The heaviest and last discovered quark was first observed at Fermilab in 1995. There are six different kinds of quarks with a wide range of masses. They are named up, down, charm, strange, top, and bottom. Quarks are the only elementary particles to experience all the known forces of nature and to have a fractional electric charge.

The interaction between quarks and gluons is responsible for almost all the perceived mass of protons and neutrons and is therefore where we get our mass.

The Higgs boson

The Higgs boson is the fundamental particle associated with the Higgs field, a field that gives mass to other fundamental particles such as electrons and quarks. A particle's mass determines how much it resists changing its speed or position when it encounters a force. Not all fundamental particles have mass. The photon, which is the particle of light and carries the electromagnetic force, has no mass at all. The Higgs boson was proposed in 1964 by Peter Higgs, François Englert, and four other theorists to explain why certain particles have mass.

Scientists confirmed its existence in 2012 through the ATLAS and CMS experiments at the Large Hadron Collider (LHC) at CERN in Switzerland. This discovery led to the 2013 Nobel Prize in Physics being awarded to Higgs and Englert. Scientists are now studying the characteristic properties of the Higgs boson to determine if it precisely matches the predictions of the Standard Model of particle physics. If the Higgs boson deviates from the model, it may provide clues to new particles that only interact with other Standard Model particles through the Higgs boson and thereby lead to new scientific discoveries.

The LHC at CERN is the highest-energy particle collider in the world. It is currently the only place scientists can create and study Higgs bosons. The DOE Office of Science (SC) contributed important accelerator magnets to help construct the LHC. DOE also supports many scientists, engineers, and technicians in the LHC program. The LHC hosts four large experimental particle detectors, two of which are partially supported by SC's Office of High Energy Physics: ATLAS and CMS. U.S. researchers account for approximately 20% and 25% of the ATLAS and CMS collaborations, respectively.

They also play leadership roles in many aspects of each experiment. These experiments are making precise measurements of the Higgs boson's properties to determine if it matches the Standard Model predictions or offers clues to new physics, exploring new particles and their interactions, and identifying the new physics of dark matter. The Higgs boson gets its mass just like other particles - from its own interactions with the Higgs field. There may be more than one Higgs boson. One theoretical model of new physics predicts five Higgs bosons.

Fundamental particles in our universe acquire mass through their interactions with the Higgs field. The Higgs boson can be a unique portal to finding signs of dark matter due to its own distinctive characteristics and properties.

Neutrinos

The neutrino is perhaps the best-named particle in the Standard Model of Particle Physics: it is tiny, neutral, and weighs so little that no one has been able to measure its mass. Neutrinos are the most abundant particles that have mass in the universe. Every time atomic nuclei come together (like in the sun) or break apart (like in a fission reactor or particle accelerator), they produce neutrinos. Even a banana emits neutrinos - they come from the natural radioactivity of the potassium in the fruit. Once produced, these ghostly particles almost never interact with other matter.

Tens of trillions of neutrinos from the sun stream through your body every second, but you can't feel them. Theorists predicted the neutrino's existence in 1930, but it took experimenters 26 years to discover the particle. Today, scientists are trying to determine the neutrino's mass, how it interacts with matter, and whether the neutrino is its own antiparticle (a particle with the same mass but opposite electric or magnetic properties) or not.

Some scientists think neutrinos might be why all antimatter (the antiparticles of all matter) disappeared after the Big Bang, leaving us in a universe made of matter. surprising - and Nobel Prize winning - discoveries. Today's investments continue that legacy through experiments and U.S.-hosted international facilities. For example, the KATRIN experiment aims to directly measure the neutrino mass.

Other experiments, including PROSPECT and the Fermilab Short-Baseline Neutrino program, use neutrinos from reactors and accelerators to search for unknown types of neutrinos beyond the three that scientists have discovered so far. Scientists are using the MAJORANA Demonstrator experiment deep under the Earth's surface to determine whether neutrinos are their own antiparticle or not.

Still other neutrino experiments, including NOvA and the upcoming international Deep Underground Neutrino Experiment, study a phenomenon called "neutrino oscillation," which looks at how mass differs between different types of neutrinos. Scientists named the three types of neutrinos they have discovered so far for the other matter particle they interact with: the electron neutrino, muon neutrino, and tau neutrino. Neutrinos change between types as they travel, something scientists call neutrino oscillation.

Particle accelerators

Particle accelerators are devices that speed up the particles that make up all matter in the universe and collide them together or into a target. This allows scientists to study those particles and the forces that shape them. Specifically, particle accelerators speed up charged particles. These are particles with a positive or negative electrical charge such as protons, atomic nuclei, and the electrons that orbit atomic nuclei. In some cases, these particles reach speeds close to the speed of light.

When the particles then collide with targets or other particles, the collisions that result can release energy, produce nuclear reactions, scatter particles, and produce other particles, such as neutrons. This gives scientists a look at what holds atoms, atomic nuclei, and nucleons together, as well as the world of particles like the Higgs boson. These particles and forces are the subject of the Standard Model of Particle Physics. Scientists also get insights into the quantum physics that govern how the world behaves at incredibly small scales.

In the quantum realm, the classical Newtonian physics that we live with every day is insufficient to explain particle interactions. How do these machines accelerate particles? They inject particles into a “beamline.” This is a pipe held at very low air pressure in order to keep the environment free of air and dust that might disturb the particles as they travel through the accelerator. The particles injected into an accelerator are referred to as a “beam.” A series of electromagnets steers and focuses the beam of particles.

In a linear accelerator or LINAC (pronounced line-ack or lin-ack), the particles shoot straight toward a fixed target. In a circular accelerator, the particles travel around a ring, constantly gaining speed. Circular accelerators can speed particles up in less overall space than a LINAC, but they tend to be more complex to build and operate. However, because the particles travel in a circle, circular accelerators create many more opportunities for particles to collide. Particles collide into each other or a target in a device called a particle detector. This device measures the particles’ speed, mass, and charge.

This information allows scientists to determine what types of particles result from a collision. The Department of Energy Office of Science supports particle accelerators and research that uses these devices primarily through its High Energy Physics and Nuclear Physics programs. These programs fund particle accelerators across the country. The Argonne Tandem Linac Accelerator System (ATLAS) at Argonne National Laboratory is the first superconducting linear accelerator for heavy ions.

At Brookhaven National Laboratory, the Relativistic Heavy Ion Collider (RHIC) is the world's first heavy ion collider, able to work with particles such as gold ions and polarized protons. Brookhaven is also home to the Accelerator Test Facility, which provides equipment to help scientists study particle acceleration and radiation generation. Fermi National Accelerator Laboratory is the core U.S. particle physics lab. Fermi was home to the world’s most powerful collider until that device, the Tevatron, was superseded by the Large Hadron Collider in 2009.

The Continuous Electron Beam Accelerator Facility at Thomas Jefferson National Accelerator Facility helps scientists study the structure of protons and neutrons. The Facility for Rare Isotope Beams at Michigan State University is a linear accelerator for studying exotic nuclei, especially isotopes. The SLAC National Accelerator Laboratory began operating a linear particle accelerator in 1962. Today, SLAC supports advanced accelerator design research for facilities around the world. Fast facts The main injector at Fermi National Accelerator Laboratory accelerates protons to 99.997 percent of the speed of light.

The largest particle accelerator in the world, the Large Hadron Collider at CERN, is more than 5 miles in diameter.

How particle physics experiments work

Matter at the smallest scale is made of elementary particles, pieces of matter that cannot be divided into anything smaller. As scientists over the past century have looked deeper and deeper into the atom, they have found the smallest things human beings have ever seen. How do they do it? The collision of particles at high energy, either with other particles or with a stationary target, allows physicists not only to look at what's inside these particles, but also to use the energy of their collisions to create different, more massive and more exotic particles of matter.

To create such high-energy collisions, scientists must use very powerful particle accelerators. To make the simplest kind of accelerator, physicists use a battery and two parallel metal plates separated by a gap. They connect one plate to the positive battery terminal and the other to the negative terminal. The battery creates an electric field in the gap between the two plates. Positively charged particles that enter the gap near the positive plate experience a force and accelerate across the gap toward the negative plate, gaining an amount of energy that depends on the voltage of the battery.

For a 10-volt battery, a proton gains 10 electronvolts, or 10 eV, as it accelerates between the plates. Putting many power supplies in a row, physicists have accelerated particles to millions of electronvolts (MeV). At some point it is impractical to increase the voltage between the metal plates, as sparks will begin to fly across the gap. To accelerate particles to even higher energy, physicists use a large number of metal plates, all with a hole in the middle. Using alternating currents, the plates can be charged either positively or negatively.

A positively charged particle, such as a proton, is drawn to the negatively charged plate in front of it, flying toward the hole. When the proton passes through the hole, the voltage of the plate is switched to a positive value, giving the proton an extra push. At the same time the next plate in front of the proton becomes negatively charged, attracting and accelerating the proton. Hence every gap between two plates provides energy to the proton as long as the voltage of the plates is switched whenever the proton crosses a hole.

In high-energy accelerators, switching the voltage happens several billion times per second, or gigahertz frequencies. Putting many plates in a row, physicists create linear accelerators, or linacs, that can accelerate charged particles to billions or trillions of electronvolts. The more plates and gaps a linac has, the higher the energy it can give to a particle - and the longer the linac gets. Rather than building longer and longer linacs, physicists are able to use magnets to guide charged particles in a circle.

In doing so, physicists are able to send the particles again and again through the same set of plates, increasing the energy of the particles with each revolution. As the particles gain energy, it is more and more difficult to keep them on the same circular path. The strength of the magnetic field must be increased. Ring-shaped particle accelerators operate the most powerful magnets in the world. The power of a ring-shaped proton accelerator is limited by its circumference and the strength of the magnets that are used. Unveiling the tiniest constituents of matter with accelerators is only half the battle.

Physicists also need extraordinary particle detectors to observe what happens in high-energy collisions. Detectors are instruments that count particles, visualize tracks, measure particle energies, record time of flight and identify different particles. Detectors can be as tiny as computer chips or as big as apartment houses, containing thousands of tons of steel and other material. Depending on the type of accelerator and the particles and forces to be studied, physicists combine various detection devices arranged in intricate configurations.

In the case of colliding beams, physicists build a detector surrounding the point at which the two beams collide. Like the layers of an onion, such a detector contains successive layers of detection devices with different functions. Close to the center, physicists place precision tracking instruments such as silicon detectors and wire chambers. Those instruments are usually surrounded by calorimeters that measure the energy of particles passing through.

The outer shell of a detector, farthest from the collision area, is devoted to detecting muons, heavy electron-like particles that can travel a couple of kilometers through rock and steel before decaying. One of the most important tasks of the detector electronics, called triggering, is the selection of collision signals that are interesting enough to be recorded permanently for later examination. It is unnecessary and impossible to keep the ten thousands of detector signals created every millionth of a second.

Physicists build and program the detector hardware to perform several levels of go or no-go decisions before passing the data to the next, more sophisticated level of processing. Physicists use detectors to analyze particles made not only in the laboratory, but in outer space. Researchers use detectors to study particles from space as they approach and enter our atmosphere in forms such as cosmic rays, gamma rays and neutrinos emitted by the sun. These experiments allow researchers to test theories about how the universe was formed, what it is made of and what its future holds.

Detectors are the product of international collaborations of physicists, all contributing their own expertise and the support of their home institutions. In return, each physicist receives access to the data recorded. To simplify the networking and data exchange within these worldwide collaborations, scientists at the European research laboratory CERN invented the World Wide Web. High-energy physics laboratories such as DESY, SLAC and Fermilab were among the first to offer Web pages in their home countries.

To analyze the enormous amount of data, particle physicists have always relied on some of the most powerful computers in the world, quickly adopting new computing technologies. The analysis of particle physics data takes place on powerful and cost-effective PC farms. Comparing simulated collision events with experimental results, sophisticated computer programs can identify the processes that took place in each collision, whether it takes place when two beams collide or at a fixed target.

Physicists use the results to test theoretical predictions, improving our knowledge of crucial parameters, contradicting theoretical expectations and discovering new phenomena.

Historical discoveries

1897: J. J. Thomson discovered the electron in a cathode-tube experiment.

1964: Gell-Mann and Zweig independently introduced the idea of quarks.

1968: Physicists at SLAC observed evidence for quarks inside the proton.

1979: Scientists at DESY reported evidence for the gluon.

1983: Physicists at CERN observed the W and Z bosons.

1995: Fermilab's CDF and DZero collaborations announced the discovery of the top quark.

2000: Fermilab announced first direct evidence for an interaction of a tau neutrino in a detector.

2012: The ATLAS and CMS collaborations independently discovered the Higgs boson at the LHC.